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HackOps: An AI-Driven Adaptive Cybersecurity Training Platform with Gamified Defense Scenarios

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Acronyms

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CTF	Capture The Flag
SQLi	SQL Injection
XSS	Cross-Site Scripting
CSRF	Cross-Site Request Forgery
LLM	Large Language Model
WAF	Web Application Firewall
OWASP	Open Web Application Security Project
IDS	Intrusion Detection System
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
TF-IDF	Term Frequency-Inverse Document Frequency
MVP	Minimum Viable Product
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, DoS, Elevation of Privilege
RBAC	Role-Based Access Control
UX	User Experience

1 Abstract

HackOps is an AI-driven cybersecurity training platform addressing defensive security education gaps through gamification and adaptive learning. Unlike traditional CTF platforms focusing on offensive skills, HackOps positions users as defenders against intelligent AI attackers. The platform implements level-based progression with SQL injection attacks, supporting expansion to XSS, CSRF, and authentication vulnerabilities. The AI engine generates dynamic payloads, adapts strategies, and escalates complexity. Through hands-on remediation, code patching, and security configuration, users develop practical defensive capabilities with intelligent feedback and adaptive difficulty adjustment.

2 Introduction

Web applications remain primary targets for cyber-attacks, with SQL injection and XSS among the most prevalent threats despite decades of awareness [1]. Traditional training emphasizes offensive security through CTF competitions, creating workforce asymmetry where professionals excel at finding vulnerabilities but lack remediation expertise [6].

2.1 Problem Statement

Current cybersecurity education faces four critical limitations: (1) platforms emphasize offensive over defensive capabilities, failing to prepare professionals for maintaining secure systems; (2) static scenarios don't adapt to learners or replicate real-world attack evolution; (3) platforms lack hands-on remediation experience beyond vulnerability identification; (4) limited intelligent feedback and adaptive difficulty progression based on individual capabilities.

2.2 Motivation

Organizations face unprecedented cyber-attacks, with SQL injection in OWASP Top 10 for two decades. Research shows AI achieves 98%+ detection rates [2,3], yet gaps remain between detection and practical defensive training. The global cybersecurity skills gap widens as organizations struggle recruiting professionals with practical defensive capabilities. AI technologies present opportunities to revolutionize training through realistic, engaging experiences.

2.3 Research Objectives

Primary objectives: (1) develop AI-powered attack systems generating diverse payloads and adapting strategies; (2) design modular level-based training architecture; (3) build interactive defense mechanisms with real-time feedback; (4) implement adaptive learning features; (5) evaluate platform effectiveness. Research questions address how AI creates realistic scenarios, which design principles engage learners, how hands-on training compares to traditional methods, optimal progression structures, and whether platforms can teach both identification and remediation.

2.4 MVP Scope

Given one-year single-student constraints, the MVP focuses on SQL injection training (Levels 1-5), basic AI attack engine with rule-based generation, core defense interface with code editing, and essential gamification with scoring and progressive difficulty.

3 Literature Review

3.1 AI in Cybersecurity

Salem et al. examined sixty studies measuring AI effectiveness in cyber threats including malware, intrusions, and injection attacks [1]. ML and deep learning combined with metaheuristic algorithms significantly improve detection versus rule-based systems. AI's ability to learn and adapt makes it valuable where adversaries constantly evolve. Deep learning revolutionized intrusion detection and malware monitoring beyond traditional ML's manual feature extraction. Research demonstrates CNNs and LSTM architectures outperform traditional systems in identifying novel attack variants and zero-day exploits [1].

3.2 SQL Injection Detection

SQL injection compromises confidentiality, integrity, and availability [2]. Multiple researchers applied ML and deep learning with remarkable success. Fathi et al.'s LSTM-based detection achieved 99.7-99.8% accuracy with perfect precision and recall [2]. Gandhi et al.'s CNN-BiLSTM model achieved 98% accuracy, demonstrating hybrid architecture effectiveness combining CNNs for feature extraction with bidirectional LSTM for learning long-term dependencies [3].

Li et al. achieved 98.35% accuracy using content matching and deep learning [4], while other research with improved TextCNN, LSTM, and attention mechanisms reached 99.57% accuracy [5]. These studies demonstrate that effective defense requires understanding multiple complementary techniques rather than single-point solutions. Deep learning models automatically learn hierarchical feature representations and scale well for large datasets with continually improving performance [5].

3.3 Cross-Site Scripting Detection

XSS remains in OWASP Top 10 for decades, requiring attention to avoid personal and financial damage [6]. Kaur et al. reviewed ML and neural network-based detection covering DNNs, decision trees, and web-log models [6]. Traditional rule-based and signature-based methods fail against sophisticated and obfuscated XSS payloads, necessitating advanced ML solutions.

Younas et al.'s LSTM-TFIDF approach combining temporal and TFIDF features with Random Forest achieved 0.99 performance [7]. Research applied Explainable AI to understand model interpretability and transparency in detecting XSS attacks. Mereani and Howe demonstrated SVM, k-NN, and Random Forests achieving 98.58-99.89% accuracy using feature sets combining language syntax and behavioral features [8]. Effective features include script tag presence, event handler patterns, JavaScript function calls, and URL structure characteristics.

Alhamyani and Alshammari evaluated Random Forest, Logistic Regression, SVMs, Decision Trees, XGBoost, and neural networks for XSS detection [9]. Models trained on real-world datasets used feature selection methods like Information Gain and ANOVA for optimization. Studies demonstrate ensemble methods combining multiple classifiers often outperform individual algorithms, suggesting defensive training should emphasize multi-layered security approaches. Deep learning research achieved 99.5% precision and 97.9% recall using LSTM and word2vec [10], demonstrating practical applicability of neural networks for web security.

3.4 Cybersecurity Training Environments

Selecting appropriate training environments is crucial for effective cybersecurity education. Chothia et al. compared CyTrONE and Labtainers containerized platforms [11], evaluating deployment complexity, resource efficiency, scalability, and educational effectiveness. CyTrONE, developed by ENISA, provides flexible frameworks for creating scenarios using Docker containers with realistic network topologies and complex multi-machine exercises. Labtainers, from Naval Postgraduate School, offers pre-configured exercises with automated assessment and detailed progress tracking.

The analysis revealed distinct trade-offs. CyTrONE offers greater flexibility for custom scenarios and scales effectively for large-scale training events, suitable for organizational programs and advanced courses [11]. However, flexibility comes with increased configuration complexity and steeper instructor learning curves. Labtainers provides structured, ready-to-deploy exercises with built-in assessment, ideal for academic environments and standardized curriculum delivery.

Vykopal et al. examined the broader ecosystem including KYPO Cyber Range Platform and applications in academic and professional contexts [12]. Their research highlights the importance of realistic, hands-on experience and demonstrates containerized environments provide effective balance between realism, safety, and resource efficiency. Modern platforms must support technical skill development, decision-making under pressure, teamwork, and incident response coordination—competencies difficult to develop through traditional lectures.

Critical design principles include: sufficient realism to transfer skills to professional practice while maintaining safety boundaries; progressive difficulty scaling from fundamental concepts to complex scenarios; immediate feedback mechanisms reinforcing learning; and scalability with resource efficiency for practical deployment [12]. These findings inform HackOps design, adopting containerization principles while focusing on defensive skills—an underserved area in existing platforms.

3.5 Gamification and Research Gaps

Research demonstrates gamification increases engagement and retention through clear objectives, immediate feedback, progressive challenges, and achievement systems. However, existing CTF platforms emphasize offensive skills, creating workforce asymmetry. Studies show personalizing difficulty and providing context-sensitive hints significantly improves outcomes.

Despite progress in AI detection and training environments, gaps exist: limited work explores AI as intelligent adversaries for training; platforms emphasize offensive over defensive skills; research on platforms like CyTrONE and Labtainers shows they excel at

realistic environments but lack focus on defensive skill development or adaptive AI adversaries [11, 12]; adaptive learning focuses on content presentation rather than dynamically modifying attack complexity; and limited research integrates multiple attack vectors in progressive environments.

HackOps addresses gaps by: inverting AI detection capabilities to generate adaptive attacks; utilizing containerization principles while focusing on defensive skills; positioning users as defenders actively remediating vulnerabilities; incorporating gamification enhancing engagement; and implementing adaptive mechanisms personalizing experiences based on performance. This integration of AI-driven attacks, proven containerization, defensive training focus, gamification, and adaptive learning represents novel contribution to cybersecurity education research.

4 Threat Modeling and Security

4.1 Threat Model and Requirements

Applying STRIDE methodology [13]: **Spoofing** (user impersonation), **Tampering** (scenario modification), **Repudiation** (logging needs), **Information Disclosure** (data exposure), **Denial of Service** (resource exhaustion), **Elevation of Privilege** (sandbox escape). Attack tree analysis [14] maps vectors including container escape and AI manipulation.

Security requirements: (1) Docker containerization with resource limits; (2) sandboxed execution; (3) multi-factor authentication and RBAC; (4) comprehensive audit trails; (5) resource quotas; (6) input validation; (7) automated scanning. Risk mitigations include hardened container configurations for escape prevention, strict network isolation, terms of service for skill misuse, and synthetic data usage. Simulated vulnerabilities have explicit boundaries ensuring learned attacks cannot directly apply to production while defensive skills transfer to real-world implementations.

5 Methodology

5.1 System Architecture

Figure 1 illustrates the architecture with AI attack engine, web application simulator, defense interface, and databases.

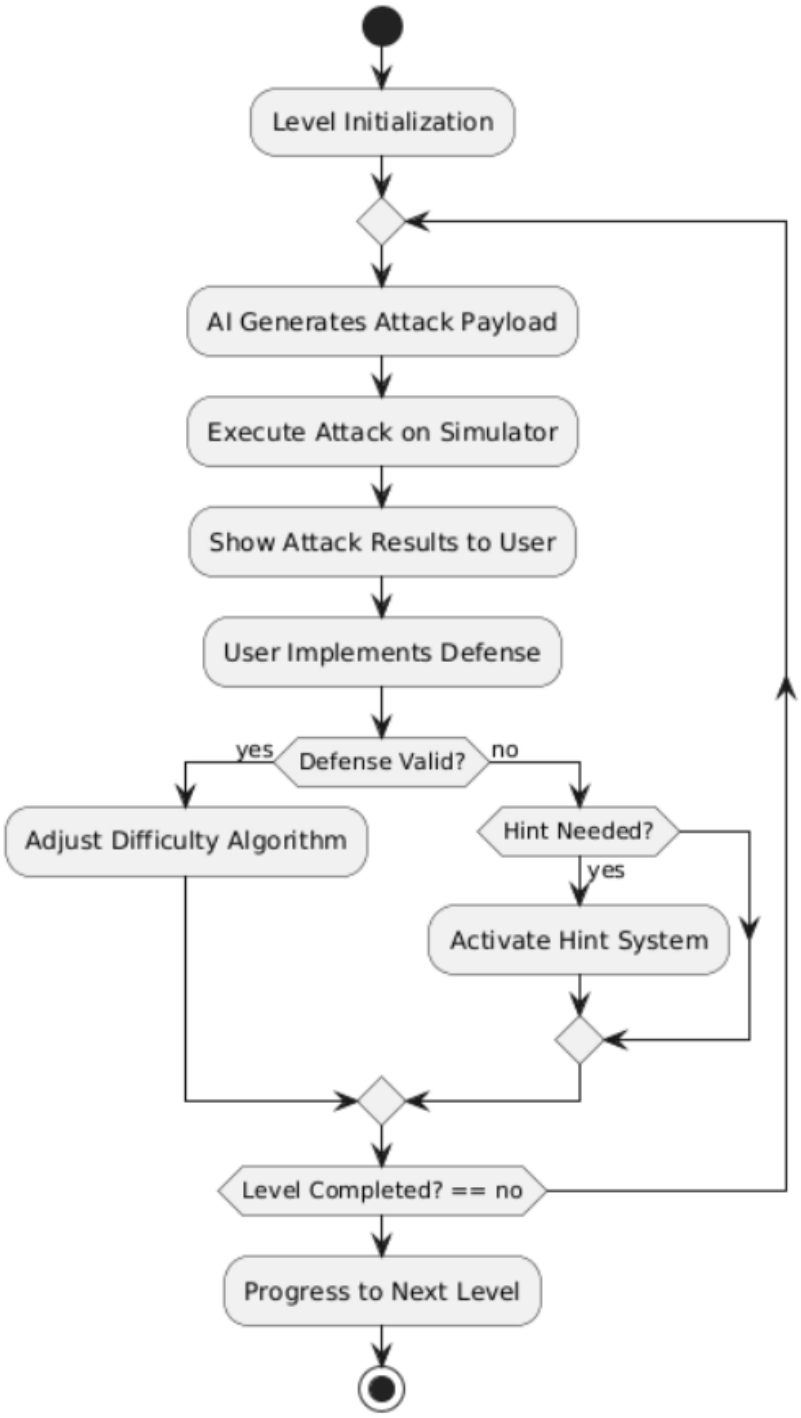

Figure 1: System Architecture Overview

The simulator includes vulnerable authentication forms, search functionality, database interfaces, and sandboxed environments. The AI engine generates SQL injection payloads using rule-based techniques and ML, analyzes responses, adapts strategies, escalates complexity, and maintains knowledge bases. The defense interface provides monitoring, code editing, security configuration, testing, and feedback.

Figure 2 shows interaction sequences during attack-defense cycles, while Figure 3 illustrates level completion workflow.

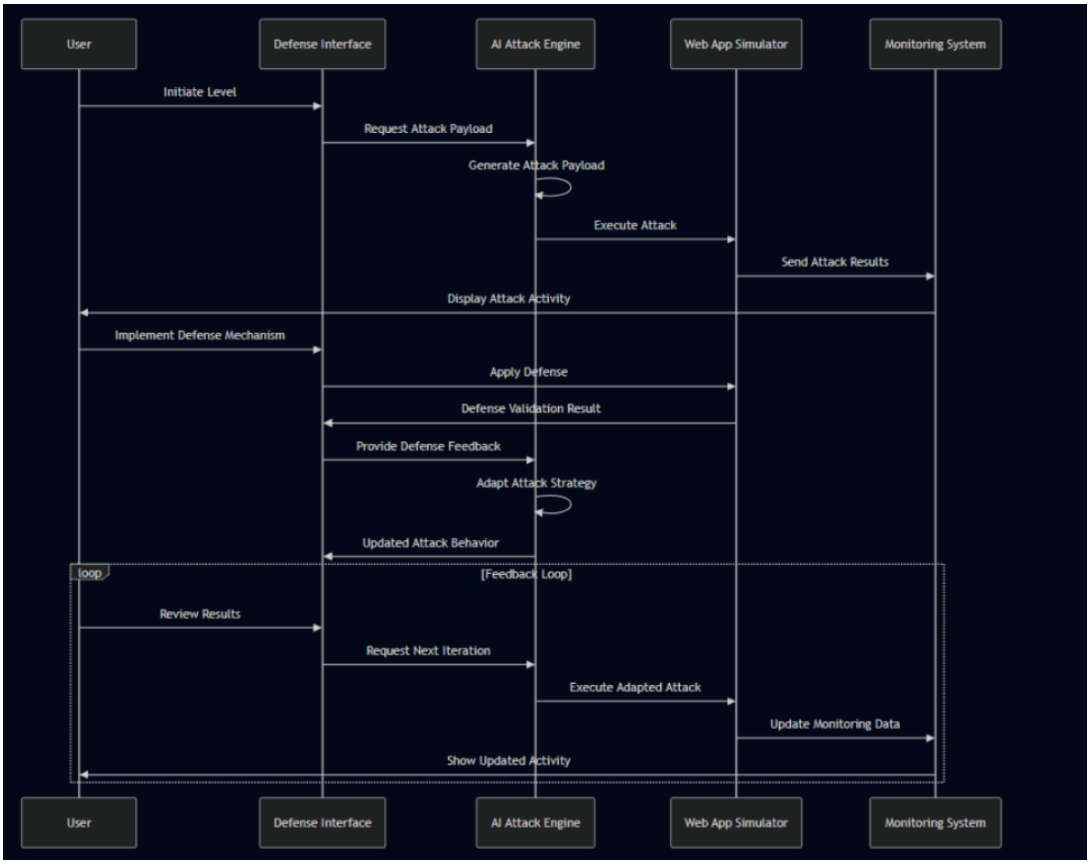

Figure 2: Attack-Defense Interaction Sequence

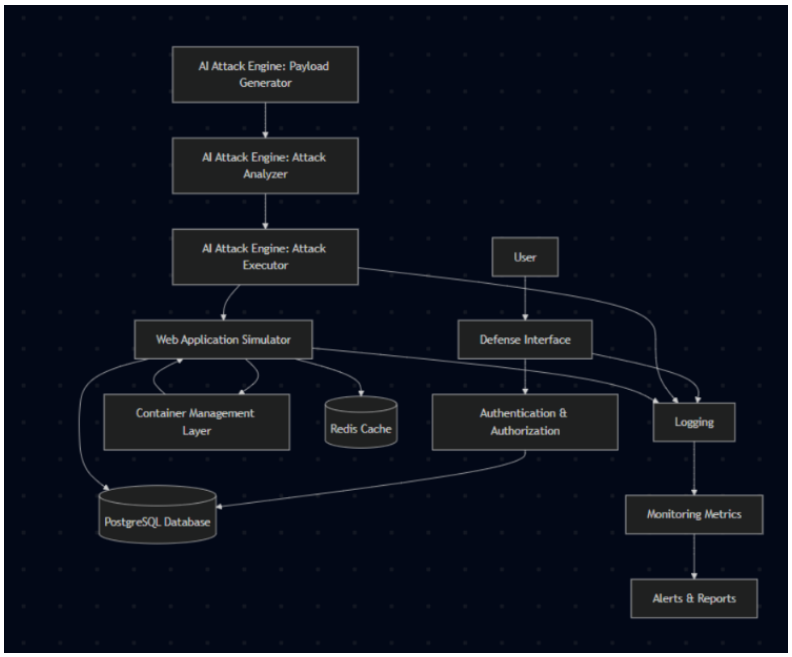

Figure 3: Level Completion Workflow

5.2 Technology Stack

Backend: Python with Flask/Django, PostgreSQL, TensorFlow/PyTorch, Redis, Docker. Frontend: React.js, Monaco Editor, D3.js, WebSocket.

5.3 Game Design

Objectives: Levels 1-3 (basic defense, block 5 attacks); Levels 4-6 (parameterized queries, 80% defense rate); Levels 7-9 (WAF configuration, sustained defense).

Scoring: 100 base points, up to 50 speed bonus, 1.0-2.0x efficiency multiplier, -10 per hint, -5 per failure, 25% streak bonus.

Rewards: Badges, unlockables, leaderboards, skill trees, certificates.

Failure: Three breaches, database compromise, 30-minute timeout, configuration crash.

Adaptation: Track performance metrics; calculate $S = (w_1 \cdot SuccessRate) + (w_2 \cdot \frac{1}{ResponseTime}) - (w_3 \cdot HintUsage)$; adjust difficulty: if $S > 0.8$ increase 20%, if $S < 0.5$ decrease 15%; AI emphasizes struggle patterns; reassess every 5 cycles.

5.4 Evaluation

Quantitative: performance scores, completion times, hint frequency, failed attempts, pre/post-assessments. Qualitative: UX surveys, interviews, strategy analysis, usability feedback. Comparative analysis against traditional methods measures retention, application, and engagement.

Project timeline available at: Gantt Chart. Phases: literature review (Months 1-2), design (Month 3), implementation (Months 4-6), AI development (Months 7-8), testing (Months 9-10), evaluation (Months 11-12).

6 Conclusion

HackOps develops an innovative AI-driven platform addressing defensive security education gaps by positioning users as defenders against intelligent AI adversaries. The modular architecture supports future expansion, demonstrating AI-powered adaptive training potential. Integration of containerization technologies from CyTrONE and Labtainers with novel AI-driven attacks creates unique training combining realistic scenarios with progressive difficulty. Comprehensive threat modeling ensures safe educational experiences.

Future work includes expanding to XSS, CSRF, and authentication vulnerabilities; enhancing AI with sophisticated models; curriculum integration; and developing mobile applications, VR interfaces, and collaborative modes for comprehensive OWASP Top 10 training.

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